FRC 2023 CHARGED UP — Top Robot Design Deep Dive

A study reference for effective CHARGED UP design & implementation, drawn from the Spectrum 3847 CAD Collection, team CAD/code releases (Onshape / GrabCAD), and each team's Chief Delphi reveal / CAD-release thread.

The CHARGED UP design landscape

The game

CHARGED UP is a cone-and-cube game: robots collect CONES and CUBES from the floor or the SINGLE/DOUBLE SUBSTATIONS and score them on the GRID's three rows (hybrid, mid, high) for links and a co-op bonus, then in the endgame DOCK and ENGAGE on the CHARGE STATION. Most elite designs paired a fast cone/cube cycle (an arm or elevator + a cone-and-cube-capable end effector) with reliable auto-balance; many top teams ran a telescoping or double-jointed arm and treated triple-balance/links as the high-value targets.

How to use this document

These are the most thoroughly-documented CHARGED UP robots from the CAD-release set, each with real subsystem detail mined from the team's Chief Delphi reveal/release thread. For every team we capture archetype, the four subsystems (shooter / intake / indexer / drivetrain), vision, public CAD availability, and 'lessons for students'. ★ marks the curated study picks with the richest reproducible documentation. EPA / rank are pending a Statbotics API outage and will be backfilled when it recovers.

Profiled robots at a glance

	Team	Robot	Local EPA	Rec	Archetype	Public CAD/Design
★	**1678 Citrus Circuits**	Tangerine Tumbler	—	—	Elite elevator cone/cube robot; full CAD + code release	CD thread
★	**971 Spartan Robotics**	Doffle	—	—	Tank-drive cone/cube robot — kept tank in a swerve-dominated y	Onshape CAD
★	**6328 Mechanical Advantage**	(2023)	—	—	Open-Alliance, software-led; reliable-and-simple philosophy; A	CD thread
★	**2910 Jack in the Bot**	Phantom	—	—	Strong PNW cone/cube robot; CAD + tech binder	CD thread
★	**1114 Simbotics**	Simbot Scizor	—	—	Design-process leaders; CAD + code + reusable base	CD thread
★	**695 Bison Robotics**	Goldmodule	—	—	Swerve cone/cube robot; public code + a swerve-CAD teaching re	CD thread
★	**3005 RoboChargers**	AMP	—	—	Cone/cube robot with a widely-praised end effector	Onshape CAD
★	**2767 Stryke Force**	Snake Eyes	—	—	Cone/cube robot with an operator-friendly vision-align indicat	Onshape CAD
★	**1706 Ratchet Rockers**	Whiplash	—	—	Cone/cube robot with a grid-registering frame	Onshape CAD
★	**118 Robonauts**	Echo	—	—	Cone/cube robot; also the FRC Everybot authors	GrabCAD

★ = recommended study pick (downloadable CAD or a full technical binder + a clear design lesson). EPA shown is Local EPA (approx), computed from The Blue Alliance match data.

Team-by-team profiles

★ Study pick	**1678 Citrus Circuits** Tangerine Tumbler USA-CA	**EPA pending**

Elite elevator cone/cube robot; full CAD + code release

SCORING (CONE / CUBE) Cone/cube end effector for grid scoring (detail in released CAD).	**INTAKE** Cone/cube intake feeding the elevator (see CAD).
ARM / ELEVATOR Elevator superstructure (cable/spool-tensioned; discussed in the release Q&A).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check code)	**PUBLIC CAD / DESIGN** CD CAD & code release

LESSONS FOR STUDENTS

- A top team's full CAD + code, with a public robot-history archive going back to 2009 — study how a program iterates year over year.
- Cable/spool elevator tensioning is a classic gotcha worth understanding from their Q&A.

CD thread: https://www.chiefdelphi.com/t/437632 · CD CAD & code release · CD reveal

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★ Study pick	**971 Spartan Robotics** Doffle USA-CA	**EPA pending**

Tank-drive cone/cube robot — kept tank in a swerve-dominated year by choice

SCORING (CONE / CUBE) Cone/cube scoring end effector (detail in released CAD).	**INTAKE** Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	**DRIVETRAIN** 6-wheel TANK / west-coast drive — a deliberate choice to keep the robot to a reasonable number of degrees of freedom while everyone else went swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A powerhouse team deliberately staying with tank drive is a great lesson in choosing simplicity / fewer DOFs over the trendy option.
- 971's mechanical execution and packaging are worth studying regardless of drivetrain.

Onshape CAD: https://frc971.onshape.com/documents/db1b739c445de530fb406dc6/v/bc9815d549b9ffd9f7251ac8/e/e81d7e62d16fb20bf281c8e4 · CD reveal — Doffle

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★ Study pick	**6328 Mechanical Advantage** (2023) USA-MA	**EPA pending**

Open-Alliance, software-led; reliable-and-simple philosophy; AdvantageKit authors

SCORING (CONE / CUBE) Cone/cube end effector (mechanism detail in released CAD).	**INTAKE** Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	**DRIVETRAIN** Swerve.
VISION Advanced logging + vision via their own AdvantageKit framework (see code).	**PUBLIC CAD / DESIGN** CD 2023 Open-Alliance build thread

LESSONS FOR STUDENTS

- 6328 author AdvantageKit — the reference FRC logging/replay framework; their code is the model for log-based development.
- Explicit 'reliable, relatively simple, lean into software strengths' design philosophy + full-team prototyping is worth emulating.

CD thread: https://www.chiefdelphi.com/t/420691 · CD 2023 build thread · Onshape CAD

★ Study pick **2910 Jack in the Bot** Phantom
USA-WA
EPA pending

Strong PNW cone/cube robot; CAD + tech binder

SCORING (CONE / CUBE) Cone/cube scoring end effector (full detail in the tech binder).	**INTAKE** Cone/cube intake (tech binder).
ARM / ELEVATOR Arm/elevator superstructure (tech binder).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check binder)	**PUBLIC CAD / DESIGN** CD CAD & tech binder release

LESSONS FOR STUDENTS

- Ships both Onshape and STEP/Parasolid plus a tech binder — a complete, clone-friendly documentation set.
- A well-documented benchmark PNW robot to compare against the elevator/arm field.

CD thread: https://www.chiefdelphi.com/t/436653 · CD CAD & tech binder

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★ Study pick **1114 Simbotics** Simbot Scizor
CAN-ON
EPA pending

Design-process leaders; CAD + code + reusable base

SCORING (CONE / CUBE) Cone/cube scoring end effector (detail in released CAD).	**INTAKE** Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check code)	**PUBLIC CAD / DESIGN** CD CAD & code release

LESSONS FOR STUDENTS

- Released the robot CAD plus Simbot-Base — a maintained starter code base with up-to-date utilities to bootstrap a season fast.
- 1114 are long-standing design-process mentors; study their packaging and conventions.

CD thread: https://www.chiefdelphi.com/t/447607 · CD CAD & code release · Code — Simbotics/2023-Simbot-Scizor

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★ Study pick **695 Bison Robotics** Goldmodule
USA-ND
EPA pending

Swerve cone/cube robot; public code + a swerve-CAD teaching resource

SCORING (CONE / CUBE) Cone/cube scoring end effector (see CAD).	**INTAKE** Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	**DRIVETRAIN** Swerve; code developed in public, conventions carried from Rapid React.
VISION Offseason vision/odometry code rewrite (see GitHub).	**PUBLIC CAD / DESIGN** CD CAD & code release

LESSONS FOR STUDENTS

- They published a parametric, template-part-based MK4i swerve SOLIDWORKS tutorial series — an outstanding learn-to-CAD-swerve resource.
- Developing code in public + documenting offseason swerve/vision rewrites is a model for continuous learning.

CD thread: https://www.chiefdelphi.com/t/446357 · CD CAD & code release · Code — FRCTeam695

★ Study pick	**3005 RoboChargers** AMP USA-TX	**EPA pending**

Cone/cube robot with a widely-praised end effector

SCORING (CONE / CUBE) A cone/cube end effector that drew a lot of community praise on reveal ('Holy end effector!') — study its grip/geometry in the released CAD.	**INTAKE** Cone/cube intake integrated with the end effector (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A standout, much-admired end effector — a great single-mechanism study for handling both cones and cubes.
- Reverse-engineer the released Onshape to see how one gripper manages two very different game pieces.

Onshape CAD: https://cad.onshape.com/documents/230339548ba1828471e00cef/w/7608c62ea34bc84ee4acccd6/e/6a002e572ec5e1812bae3918 · CD reveal — AMP

★ Study pick	**2767 Stryke Force** Snake Eyes USA-MI	**EPA pending**

Cone/cube robot with an operator-friendly vision-align indicator

SCORING (CONE / CUBE) Cone/cube scoring end effector (see CAD).	**INTAKE** Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	**DRIVETRAIN** Swerve.
VISION Camera auto-align to the grid, with a light-up indicator that turns on when the camera alignment engages — a clean driver-feedback idea.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A visible status light tied to vision auto-align is a simple, high-value human-factors touch worth copying.
- Study how the alignment indicator integrates with the scoring cycle.

Onshape CAD: https://cad.onshape.com/documents/04a54b2035c31e128e1dfced/w/505caf917c2302fdb0b5bdd9/e/4c7eedc3e2c9f9a4836827bf · CD reveal — Snake Eyes

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★ Study pick	**1706 Ratchet Rockers** Whiplash USA-MO	**EPA pending**

Cone/cube robot with a grid-registering frame

SCORING (CONE / CUBE) Cone/cube scoring end effector (see CAD).	**INTAKE** Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A slightly OCTAGONAL frame perimeter lets them jam the bumper into the grid to register/line up for scoring — a clever passive-alignment trick.
- Study how frame geometry can do alignment work that would otherwise need sensors.

Onshape CAD: https://cad.onshape.com/documents/80bfb649a2c78e4c6edec3d3/w/6b74b6511ba0a48d4efc8200/e/9dac5cbb17ec4689a721dff4 · CD reveal — Whiplash

★ Study pick	**118 Robonauts** Echo USA-TX	**EPA pending**

Cone/cube robot; also the FRC Everybot authors

SCORING (CONE / CUBE) Cone/cube scoring end effector (detail in the released GrabCAD).	**INTAKE** Cone/cube intake (see CAD).
ARM / ELEVATOR Arm/elevator superstructure (see CAD).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** GrabCAD — Echo

LESSONS FOR STUDENTS

- 118 lead the Everybot program — the canonical low-cost, accessible CHARGED UP design many rookies build.
- Compare their full comp robot 'Echo' against the simplified Everybot to see design trade-offs.

GrabCAD: https://grabcad.com/library/robonauts-118-2023-robot-echo-1 · GrabCAD — Echo · CD reveal video

Additional well-documented robots for study

Each has a strong public release (CAD, code, binder, or detailed reveal) and a clear design lesson. ★ marks the recommended study picks.

Sources: the Spectrum 3847 CAD Collection and team CAD releases (Onshape, GrabCAD), Chief Delphi reveal / CAD-release threads, and match data from The Blue Alliance. EPA shown is Local EPA (approx), independently computed — not official Statbotics. "Unconfirmed" means a detail was not publicly published, not that it is absent.